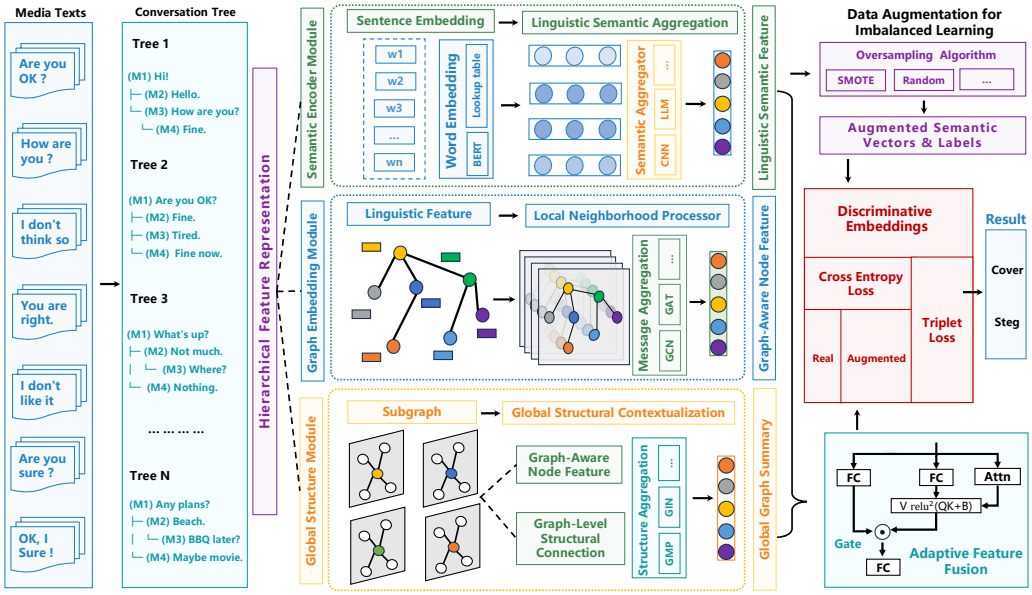


Figure 2: Our proposed GSDFuse architecture. This multi-stage pipeline is designed to identify cognitive inconsistencies by effectively aggregating multi-dimensional weak signals in social media steganalysis, leveraging diverse feature representations and optimized learning strategies.

like oversampling to mitigate the effects of sparse steganographic instances. Subsequently, the **Optimization for Discriminative Embeddings** module utilizes specialized loss functions. Its objective is to sculpt an embedding space where steganographic samples are maximally separable from benign ones, directly enhancing the model's ability to detect faint and ambiguous signals.

This multi-stage, modular architecture provides a comprehensive approach to steganalysis and offers flexibility for future enhancements and adaptations within each component to tackle evolving steganographic techniques.

## 3.4 Detailed Components

### 3.4.1 Hierarchical Feature Representation

**Semantic Node Profiling** Each message node within the dialogue tree, denoted as a sentence $x$, is initially represented as a sequence of token identifiers $x = (w_1, w_2, \ldots, w_L)$, where $L$ is the length of the sentence. These discrete token IDs are first mapped to dense vector representations through an optimizable embedding lookup table $E$, resulting in a sequence of token embeddings. Subsequently, a Semantic Composition Aggregator (SCA) module processes these token embeddings to produce a fixed-length semantic vector $\mathbf{s}_x \in \mathbb{R}^{d_s}$ that encapsulates the core meaning of the individual message. This process can be formally expressed as:

$$\mathbf{s}_x = \mathrm{SCA}(E(w_1), E(w_2), \ldots, E(w_L)). \quad (1)$$

Various models can serve as the SCA; for example, Convolutional Neural Networks (CNNs) capture n-gram linguistic correlations, an approach demonstrated in TS-CSW (Yang et al., 2020b).

**Topological Contextualization** To incorporate structural information from the dialogue tree $G = (V, A)$, where $V$ denotes message nodes and $A$ the adjacency matrix of reply-to relationships, we employ GNNs. These networks iteratively refine a node's representation by processing information from its local neighborhood. Starting with an initial node representation $\mathbf{h}_v^{(0)} = \mathbf{s}_v$ , GNNs operate in layers. At each layer $k$, an aggregated message $\mathbf{m}_{\mathcal{N}(v)}^{(k)}$ is first computed from the representations of node $v$'s neighbors $\mathcal{N}(v)$:

$$\mathbf{m}_{\mathcal{N}(v)}^{(k)} = \mathrm{AGG}^{(k)}\left(\{\mathbf{h}_u^{(k-1)} : u \in \mathcal{N}(v)\}\right). \quad (2)$$

This message is then used to update node $v$:

$$\mathbf{h}_v^{(k)} = \mathrm{UPDATE}^{(k)}\left(\mathbf{h}_v^{(k-1)}, \mathbf{m}_{\mathcal{N}(v)}^{(k)}\right), \quad (3)$$

where $\mathrm{AGG}^{(k)}$ and $\mathrm{UPDATE}^{(k)}$ are layer-specific aggregator and update functions.

To effectively capture fine-grained structural distinctions critical for steganalysis, we can leverage powerful GNN architectures. For instance, Graph Isomorphism Network (GIN) (Xu et al., 2018) is known for its strong discriminative power, approaching the theoretical limit of the Weisfeiler-Lehman (WL) test for graph isomorphism. A GIN

layer updates node representations using a sum aggregator and a Multi-Layer Perceptron (MLP):

$$\mathbf{a}_v^{(k)} = (1 + \epsilon^{(k)}) \cdot \mathbf{h}_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(k-1)}. \quad (4)$$

This aggregated representation $\mathbf{a}_v^{(k)}$ is then transformed by a Multi-Layer Perceptron (MLP) specific to the $k$-th layer:

$$\mathbf{h}_v^{(k)} = \text{MLP}^{(k)} \left( \mathbf{a}_v^{(k)} \right), \quad (5)$$

where $\epsilon^{(k)}$ is a learnable parameter or a fixed scalar. Through stacking such layers, the GNN produces topologically-aware node embeddings $\mathbf{h}_v$ that reflect both semantic content and structural context within the dialogue tree.

#### 3.4.2 Data Augmentation for Imbalanced Learning

A primary challenge in steganalysis is the inherent steganographic signal sparsity, where instances of steganography (stego) are significantly outnumbered by benign cover messages. This pronounced class imbalance biases models towards the majority class, hampering their ability to detect the subtle cognitive inconsistencies introduced by advanced steganographic techniques. To address this, we employ data augmentation strategies, specifically oversampling techniques, to rebalance the training distribution and enhance the model's exposure to minority class characteristics.

One widely adopted method is the Synthetic Minority Over-sampling Technique. SMOTE (Sun et al., 2024) operates by generating new synthetic minority samples in the feature space. For a given minority sample $\mathbf{x}_i$, a new sample $\mathbf{x}_{\text{new}}$ is created by interpolating between $\mathbf{x}_i$ and one of its randomly selected $k$-nearest minority neighbors, $\mathbf{x}_j$:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_j - \mathbf{x}_i), \quad (6)$$

where $\lambda$ is a random scalar in $[0, 1]$. By applying SMOTE, typically within the learned embedding space to ensure semantic coherence of synthetic samples, we enrich the training set with diverse yet plausible stego instances.

#### 3.4.3 Cross-Modal Feature Integration

Detecting steganography in social media requires effective aggregation of multi-dimensional weak signals, which are often subtly dispersed. Naive fusion methods (e.g., concatenation or averaging) for features like semantic profiles $\mathbf{s}_v$, topological embeddings $\mathbf{h}_v$, and global structure summaries $\mathbf{g}_v$ often fail to resolve these nuanced and potentially conflicting indicators.

To address this, our method employs Gated Attention Unit (GAU) (Hua et al., 2022) for **Adaptive Feature Fusion**. Given the concatenated multi-modal feature vector for a node $v$, $\mathbf{x}_{\text{concat},v} = [\mathbf{s}_v; \mathbf{h}_v; \mathbf{g}_v]$, where $[\cdot]$ denotes concatenation, the GAU produces an integrated feature $\mathbf{f}_v$:

$$\mathbf{f}_v = \text{GAU}(\mathbf{x}_{\text{concat},v}). \quad (7)$$

The GAU utilizes learnable gating and attention mechanisms to dynamically weigh input features and interactions per instance. This allows it to effectively arbitrate how semantic, local topological, and global structural information is combined. Adaptive fusion is crucial for the aggregation of multi-dimensional weak signals to identify subtle cognitive inconsistencies indicative of steganography, when signals are conflicting or dispersed.

#### 3.4.4 Optimization for Discriminative Embeddings

Despite adaptive fusion, feature embeddings $\mathbf{f}_v$ of steganographic (stego) and benign messages may remain close, especially when subtle cognitive inconsistencies arise from multi-dimensional weak signals. To enhance separability, we optimize for highly discriminative embeddings using a specialized loss function.

Our approach incorporates Triplet Loss (Schroff et al., 2015), a metric learning technique. It structures the embedding space by pulling same-class samples together and pushing different-class samples apart. For an anchor $\mathbf{f}_a$ (e.g., cover), a positive $\mathbf{f}_p$ (e.g., another cover), and a negative $\mathbf{f}_n$ (e.g., stego), Triplet Loss aims to satisfy:

$$D(\mathbf{f}_a, \mathbf{f}_p) + \alpha < D(\mathbf{f}_a, \mathbf{f}_n), \quad (8)$$

where $D(\cdot, \cdot)$ is a distance function and $\alpha$ is a predefined margin. The loss is:

$$L_{\text{triplet}} = \max(0, D(\mathbf{f}_a, \mathbf{f}_p) - D(\mathbf{f}_a, \mathbf{f}_n) + \alpha), \quad (9)$$

where the margin $\alpha$ compels the model to distinguish subtle steganographic traces. This focus on hard-to-separate pairs improves sensitivity.

Triplets are selected via semi-hard negative mining. For both discriminative embeddings and accurate classification, $L_{\text{triplet}}$ is combined with Cross-Entropy $L_{\text{CE}}$:

$$L_{\text{total}} = L_{\text{CE}} + \beta \cdot L_{\text{triplet}}, \quad (10)$$

Table 1: F1 scores of steganalysis methods on the Sina, Tweet, and Reddit datasets using classic steganography

| Algorithm | | HC | | | | | | AC | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | | Sina | | Tweet | | Reddit | | Sina | | Tweet | | Reddit | |
| BPW | | 1.00 | 2.59 | 1.00 | 4.35 | 2.63 | 3.42 | 2.00 | 3.27 | 5.68 | 6.94 | 0.45 | 4.69 |
| SRS=10% | TS-ATT[IWDW'21] | 65.78 | 21.39 | 68.41 | 70.89 | 84.79 | 80.93 | 36.46 | 14.86 | 64.69 | 44.02 | 88.88 | 72.88 |
| | CATS[ICONIP'23] | 51.07 | 15.63 | 68.44 | 60.27 | 75.13 | 68.18 | 30.96 | 21.89 | 56.09 | 49.29 | 86.77 | 62.22 |
| | TGCA[ICASSP'25] | 63.16 | 31.78 | 70.26 | 71.17 | 86.52 | 81.08 | 43.08 | 24.38 | 65.64 | 52.20 | 89.05 | 73.23 |
| | **ours** | **73.24** | **46.13** | **73.84** | **72.16** | **86.86** | **82.65** | **55.23** | **36.93** | **67.48** | **55.35** | **90.86** | **75.33** |
| SRS=20% | TS-ATT[IWDW'21] | 68.24 | 47.75 | 79.21 | 77.37 | 90.25 | 84.20 | 55.17 | 24.79 | 70.34 | 61.35 | 91.89 | 76.05 |
| | CATS[ICONIP'23] | 65.01 | 47.91 | 80.27 | 74.25 | 86.16 | 76.67 | 53.59 | 36.06 | 66.78 | 58.51 | 91.87 | 66.80 |
| | TGCA[ICASSP'25] | 71.67 | 54.05 | 81.06 | 76.88 | 90.32 | 85.13 | 57.90 | 41.87 | 72.69 | 67.03 | 91.46 | 78.81 |
| | **ours** | **80.10** | **60.46** | 80.84 | **78.80** | **90.71** | **85.30** | **67.28** | **45.03** | 72.28 | **71.64** | **92.18** | 77.66 |
| SRS=30% | TS-ATT[IWDW'21] | 75.40 | 59.07 | 86.28 | 81.49 | 90.91 | 85.72 | 55.20 | 33.28 | 77.60 | 66.45 | 91.51 | 82.45 |
| | CATS[ICONIP'23] | 75.95 | 59.10 | 86.41 | 78.43 | 85.98 | 79.80 | 62.09 | 48.11 | 78.38 | 68.52 | 90.61 | 78.30 |
| | TGCA[ICASSP'25] | 77.62 | 64.35 | 87.61 | 82.83 | 91.13 | 86.42 | 60.40 | 56.11 | 79.76 | 68.45 | 92.30 | 82.86 |
| | **ours** | **83.40** | **73.23** | **87.72** | **83.01** | **91.24** | **86.92** | **75.54** | **64.54** | **80.59** | **72.18** | **92.56** | **84.72** |
| SRS=40% | TS-ATT[IWDW'21] | 82.89 | 58.71 | 90.36 | 87.08 | 92.64 | 88.68 | 67.33 | 40.41 | 84.93 | 71.72 | 93.18 | 84.44 |
| | CATS[ICONIP'23] | 83.66 | 72.84 | 90.85 | 84.19 | 89.61 | 85.35 | 68.53 | 62.79 | 82.65 | 76.37 | 93.27 | 81.38 |
| | TGCA[ICASSP'25] | 85.92 | 73.04 | 91.98 | 88.53 | 93.09 | 88.99 | 72.44 | 65.77 | 85.65 | 77.09 | 94.13 | 86.49 |
| | **ours** | **88.80** | **79.64** | 91.58 | 87.18 | **93.10** | **89.43** | **79.72** | **66.43** | **86.13** | 76.94 | **94.19** | **87.37** |
| SRS=50% | TS-ATT[IWDW'21] | 84.76 | 63.97 | 93.01 | 89.43 | 92.64 | 92.07 | 69.97 | 44.08 | 87.88 | 81.04 | 95.31 | 88.15 |
| | CATS[ICONIP'23] | 87.66 | 79.28 | 93.39 | 88.16 | 90.77 | 88.61 | 81.03 | 73.77 | 87.51 | 80.32 | 95.22 | 85.75 |
| | TGCA[ICASSP'25] | 88.18 | 77.79 | 94.12 | 90.45 | 93.64 | 92.51 | 77.41 | 73.27 | 88.67 | 83.75 | 96.09 | 88.98 |
| | **ours** | **91.91** | **84.36** | **94.30** | **90.85** | **94.59** | **92.93** | **83.98** | **76.57** | **89.23** | **85.04** | **96.17** | **90.76** |

where $\beta$ balances the loss components.

# 4 Experiments

## 4.1 Dataset Construction and Description

We evaluate GSDFuse on Stego-Sandbox (Yang et al., 2022), a public social media linguistic steganalysis dataset. It offers dialogue from Twitter, Reddit, and Sina Weibo. It features steganography from three algorithms: Huffman Coding (HC), Arithmetic Coding (AC), and Adaptive Dynamic Grouping (ADG). Steganographic payload is measured by average Bits Per Word (BPW). Stego-Sandbox provides platform-specific sub-datasets, each split into training, validation, and test sets at approximately a 7:1:1 ratio. A key feature is its simulation of varying steganographic ratios via Sparsity Ratios of Stegos (SRS), ranging from 10% to 50%. SRS indicates the percentage of eligible texts that are steganographic. Detailed statistics can be Appendix A.1.

## 4.2 Experimental Setup

Experiments ran on 10 NVIDIA L20 GPUs, taking about 2 days. Each individual model was trained for a maximum of 200 epochs. An early stopping mechanism, with a patience of 20 epochs based on the F1-score on the validation set, was employed to select the optimal model checkpoint. Figure 3 shows how average training time per epoch varies with SRS across the datasets. We use the F1-score as our metric, ideal for imbalanced detection tasks with sparse stego data. Model performance

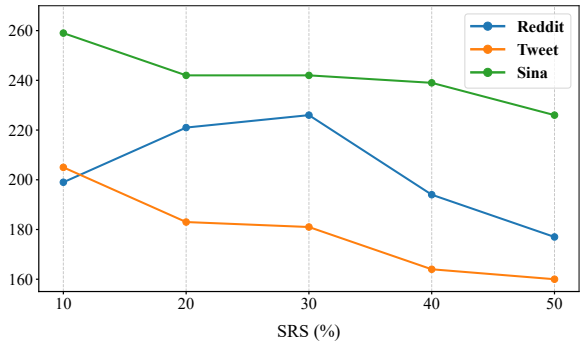


Figure 3: Average training time (s) per epoch as a function of SRS for the Reddit, Tweet, and Sina datasets.

was tested with Sparsity Ratios of Stegos (SRS) from 10% to 50%. Key model components include SMOTE over-sampling during training. GAU for adaptive feature fusion, GIN for global graph structure capture, and Triplet Loss as an auxiliary loss for more discriminative embeddings. Mini-batch subgraphs were sampled via a random walk strategy (Zeng et al., 2020). The best model checkpoint was chosen by F1-score on a validation set. All reported F1-scores are averages from 3 runs. Full component-specific hyperparameter details for all components are in Appendix B.

## 4.3 Baseline Methods

We compared our proposed method with four representative linguistic steganalysis baselines: TS-ATT (Zou et al., 2020), LSTSN (Yang et al., 2022), CATS (Pang et al., 2023), and TGCA (Lu et al., 2025). TS-ATT uses LSTM and attention for lo-